

Example:

3 Solution w/ 10% HCl → hydrogen chloride  
 20%        "-  
 40%        "-

Q: 100 l w/ 25% HCl. ← is this possible?  
 if yes, how?

A:  $x_1$  is the amount of solution 1,  $x_2$  of 2,  $x_3$  of 3

$$\left. \begin{aligned} x_1 + x_2 + x_3 &= 100 \\ \frac{10}{100}x_1 + \frac{20}{100}x_2 + \frac{40}{100}x_3 &= \frac{25}{100} \cdot 100 \end{aligned} \right\} \text{Mathematical model.}$$

→ Linear equations

$$\underbrace{a_1}_{\text{coefficients}} x_1 + \underbrace{a_2}_{\text{unknowns}} x_2 + \dots + \underbrace{a_n}_{\text{unknowns}} x_n = b$$

$x_1, x_2, x_3$  : unknowns

Linear system: System of linear equations

Basic Questions in Linear Algebra

1. Does there exist a solution?
2. If yes, do we have a unique solution or more?

Solution

unknowns:  $x_1, x_2, \dots, x_n$

Solution:  $s_1, s_2, \dots, s_n \leftarrow \text{numbers}$

Solution Set: [set of all solutions]

Important Fact

A linear system has 0, 1, or  $\infty$  many solutions.

(1)

$$x_1 + x_2 + x_3 = 100$$

$$\frac{10}{100}x_1 + \frac{20}{100}x_2 + \frac{40}{100}x_3 = 25 \quad (2)$$

(1)  $x_1 = 100 - x_2 - x_3$  substitute into (2) :

$$\frac{10}{100}(100 - x_2 - x_3) + \frac{20}{100}x_2 + \frac{40}{100}x_3 = 25 \Leftrightarrow$$

$$10 - \frac{10}{100}x_2 - \frac{10}{100}x_3 + \frac{20}{100}x_2 + \frac{40}{100}x_3 = 25 \Leftrightarrow$$

$$\cancel{\frac{10}{100}}x_2 + \cancel{\frac{30}{100}}x_3 = 15 \Leftrightarrow$$

$$x_2 + 3x_3 = 150 \Leftrightarrow x_2 = 150 - 3x_3 \quad (3)$$

$$(1) \rightarrow x_1 = 100 - 150 + 3x_3 - x_3 = -50 + 2x_3 \quad (4)$$

Quantities of litres cannot be negative

$$x_1, x_2, x_3 \geq 0$$

$$x_1 \geq 0 \Leftrightarrow -50 + 2x_3 \geq 0$$

$$x_2 \geq 0 \Leftrightarrow 150 - 3x_3 \geq 0$$

$$\boxed{\begin{array}{l} x_3 \geq 25 \\ x_3 \leq 50 \end{array}}$$

Any  $25 \leq x_3 \leq 50$  implies a solution. So we have infinitely many solutions.

Consistent (linear system): if solution exists

Inconsistent: if no solution exists.

Question:  $3x_1 + x_2 = 1$   
 $2x_1 + x_2 = 0$   
 correspond to Ques.

Solution:  $x_1 = 1$   
 $x_2 = -2$

Substitution

$$x_1 = \frac{1-x_2}{3}$$

$$2 \frac{1-x_2}{3} + x_2 = 0$$

$$\frac{2}{3} - \frac{2}{3}x_2 + x_2 = 0$$

$$\frac{2}{3} + \frac{1}{3}x_2 = 0 \Rightarrow$$

$$\cancel{\frac{1}{3}}x_2 = -\cancel{\frac{2}{3}}$$

$$x_2 = -2 \Rightarrow x_1 = 1$$

Compact way to represent linear systems

→ Coefficient Matrix

row →  $\begin{bmatrix} 1 & 1 & 1 \\ 10/100 & 20/100 & 40/100 \end{bmatrix}$   $2 \times 3$  matrix

→ Augmented Coefficient Matrix

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 100 \\ 10/100 & 20/100 & 40/100 & 25 \end{array} \right]$$

if and only if

Important  
Fact 2

A linear system is consistent  $\Leftrightarrow$   
 last column of the augmented coefficient matrix  
 in echelon form does not have a pivot.

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 100 \\ 10/100 & 20/100 & 40/100 & 25 \end{array} \right] (\times 10)$$

$$x_1 + x_2 + x_3 = 100$$

$$\frac{10}{100}x_1 + \frac{20}{100}x_2 + \frac{40}{100}x_3 = 25$$

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 100 \\ 1 & 2 & 4 & 250 \end{array} \right] \quad (2)-(1)$$

$$x_1 + x_2 + x_3 = 100$$

$$x_1 + 2x_2 + 4x_3 = 250$$

$$\left[ \begin{array}{ccc|c} \boxed{1} & 1 & 1 & 100 \\ 0 & \boxed{1} & 3 & 150 \end{array} \right]$$

$$x_1 + x_2 + x_3 = 100$$

$$x_2 + 3x_3 = 150$$

pivots: first non-zero element of a non-zero row

Matrix in echelon form:

- non-zero rows are above the zero-rows
- pivot of a row is always to the right of the pivot of the row above it.

Matrix in reduced echelon form:

- it is in echelon form
- Each pivot equals one and is the only non-zero entry in its column.

Elementary row operations

- Adding a multiple of a row to another
- Multiply a row with number  $\neq 0$
- Interchange rows

Any matrix can be transformed into a (reduced) echelon form by elementary row operations

reduced echelon form is unique  
echelon form not.

$$\left[ \begin{array}{ccc|ccc} \boxed{1} & 1 & 1 & 1 & 0 & 0 \\ 0 & \boxed{1} & 3 & 1 & 5 & 0 \end{array} \right] \xrightarrow{(1)-(2)} \left\{ \begin{array}{l} x_1 + x_2 + x_3 = 100 \\ x_2 + 3x_3 = 150 \end{array} \right.$$

$$\left[ \begin{array}{ccc|ccc} 1 & 0 & -2 & -50 & 0 & 0 \\ 0 & 1 & 3 & 150 & 0 & 0 \end{array} \right] \leftarrow \text{reduced echelon form}$$

$$\left. \begin{array}{l} x_1 - 2x_3 = -50 \\ x_2 + 3x_3 = 150 \end{array} \right\} \Rightarrow \left. \begin{array}{l} x_1 = 2x_3 - 50 \\ x_2 = 150 - 3x_3 \\ x_3 = x_3 \end{array} \right\} \begin{array}{l} \text{basic variables.} \\ \text{expressed in terms} \\ \text{of the free variable} \end{array}$$

$\uparrow$   
free variable

parametric description  
of the solution set.

Matrices equivalent (row-equivalent) if we can transform one to another w/ elementary row operations.

Example

$$\left. \begin{array}{l} 3x_1 + x_2 = 1 \\ 6x_1 + 2x_2 = 3 \end{array} \right\}$$

$$\begin{bmatrix} 3 & 1 & : & 1 \\ 6 & 2 & : & 3 \end{bmatrix} \xrightarrow{x_2} \sim \begin{bmatrix} 6 & 2 & : & 2 \\ 6 & 2 & : & 3 \end{bmatrix} \xrightarrow{(2)-(1)} \sim$$

$$\begin{bmatrix} 6 & 2 & : & 2 \\ 0 & 0 & : & 1 \end{bmatrix}$$

( by substitution:  $x_1 = \frac{1-x_2}{3}$

and  $2 - 2x_2 + 2x_2 = 3 \Leftrightarrow 2 = 3 \quad \text{⚡}$

$x_1 = ?$

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | . | . | . | . | . |
| 0 |   |   |   |   |   |
| 0 |   |   |   |   |   |
| 0 |   |   |   |   |   |